V Challenge Team Report: Phase 2

1. Team Information

- **Team Name:** Team Evolution
- **Team Members:**
    - Akaash S S
    - Om Jokhir Athitya
    - Sonavadekar Rajvardhan Sarjerao
    - Aditya Kumar
    - Anjali Yadav
- **Mentor Name & Affiliation:**
    - Dr. Poonam Kumari
    - Professor, Department of Mechanical Engineering, Indian Institute of Technology Guwahati

2. System Architecture & Control Strategy

- **System Overview:** For Phase 2, we developed a multi-agent simulation architecture to handle complex, event-driven scenarios. The system operates within a **Docker Engine**, ensuring a consistent and isolated environment for all components. The core of our architecture consists of:
    - **Car Agent:** An FMU, developed in Simulink, containing the vehicle's supervisory control logic. It manages the vehicle's state (driving, parking, charging) and tells to execute the necessary maneuvers by communicating with the bridge.
    - **Station Agent:** A separate process that simulates the charging station's state, managing availability and the charging process itself.
    - **Communication Bridge:** A middleware component responsible for facilitating real-time data exchange between the CarMaker simulation, the Car Agent, and the Station Agent.
- **Control Logic:** The Car Agent's logic is built around a state machine designed to autonomously navigate the challenge scenario. It overrides the vehicle's standard cruising behavior when specific events are detected. The primary states include:

1. **Route Following:** Standard operation, optimizing for speed and efficiency.
2. **Parking Maneuver:** Initiated upon detecting a trigger from the Station Agent.

3. **Charging:** A wait state where the vehicle is stationary and communicates with the Station Agent.

4. **Exit Maneuver:** A pre-defined sequence to safely re-join the primary route after charging is complete.

3. System Interface & Parameters

- **Interface Method:** High-frequency, real-time data exchange between the CarMaker simulation environment and our external Car Agent FMU was achieved using **Direct Variable Access (DVA)**. This provided the low-latency communication necessary for stable vehicle control.

- **Key Interface Signals:**

Signal Name	Signal Source / Destination	Purpose
Trigger_ID	DM.TriggerPoint.Id	Detects proximity to the charging station to initiate parking.
Route_Dist	Route.DistToManeuver	Provides distance information for smooth maneuver transitions.
Cur_SoC	PT.BCU.BattHV.SOC	Monitors battery state to determine when charging is complete.
Motor_Trq_Target	PT.Control.Motor.Trq_trg	Output from the FMU to control vehicle motion.

- **Maneuver Parameters:**

Parameter Name	Value	Description
Parking_Maneuver_Speed	5 km/h	Target speed for safely entering the charging bay.
Exit_Maneuver_Speed	10 km/h	Target speed for the initial phase of re-joining the route.
Exit_Maneuver_Time	4000 ms	The duration for the smooth transition back to the main route.

4. Simulation Execution & Key Challenges

- **Scenario Execution:** The simulation successfully demonstrated the agent-based logic. The vehicle followed its primary route, correctly identified the charging station trigger, executed the automated parking maneuver, waited for the charge to complete, and initiated the exit sequence.

- **Primary Challenge & Solution: Vehicle Instability During Route Re-entry** A critical challenge arose during the "Driving out of pit" maneuver. Initially, the vehicle experienced severe instability, resulting in a recurring **"Vehicle Leaves Road"** error.
    - **Root Cause Analysis:** We traced the issue to an overly aggressive maneuver definition in CarMaker. The transition profile was set to a **Step** command, which demanded an instantaneous, physically impossible change in the vehicle's target path.
    - **Solution:** The control strategy was refined by replacing the Step command with a smooth **Sine** transition, executed over a Sync. time of 4 seconds. This crucial adjustment provided a realistic steering profile, stabilized the vehicle dynamics, and allowed for a seamless and safe re-entry into traffic.

5. Team Experience and Learning

Phase 2 of the EV Challenge was an exceptional learning experience that pushed us beyond algorithm design into the realm of complex system integration. Developing and debugging the multi-agent architecture taught us the importance of robust inter-process communication and state management. The "Vehicle Leaves Road" error was a critical lesson in vehicle dynamics, emphasizing the need to translate control logic into physically realistic commands. This project provided invaluable hands-on experience in solving real-world autonomous driving challenges, from high-level mission planning down to the nuances of dynamic maneuver execution.

6. Optional Attachments

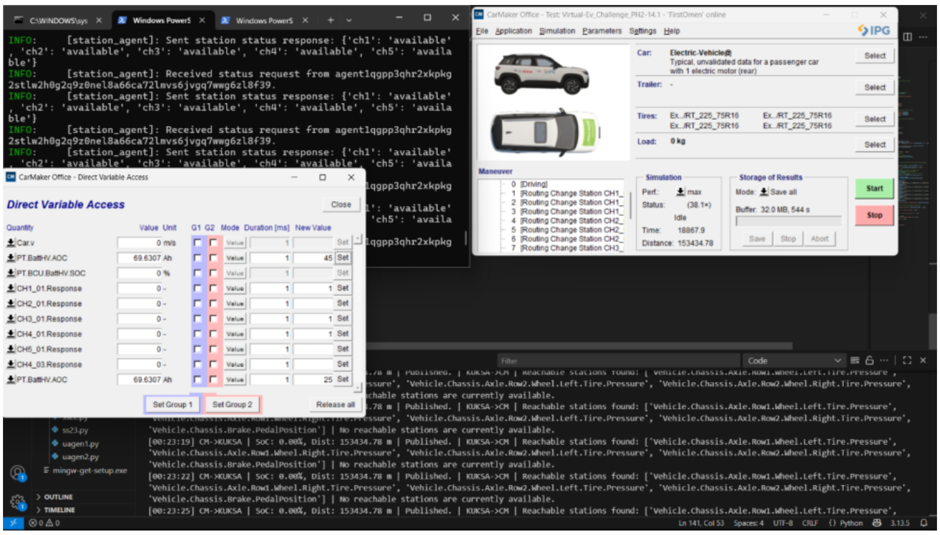


Fig 1. Multi-Agent System Architecture Diagram

Fig 2. Pycarmaker Bridge

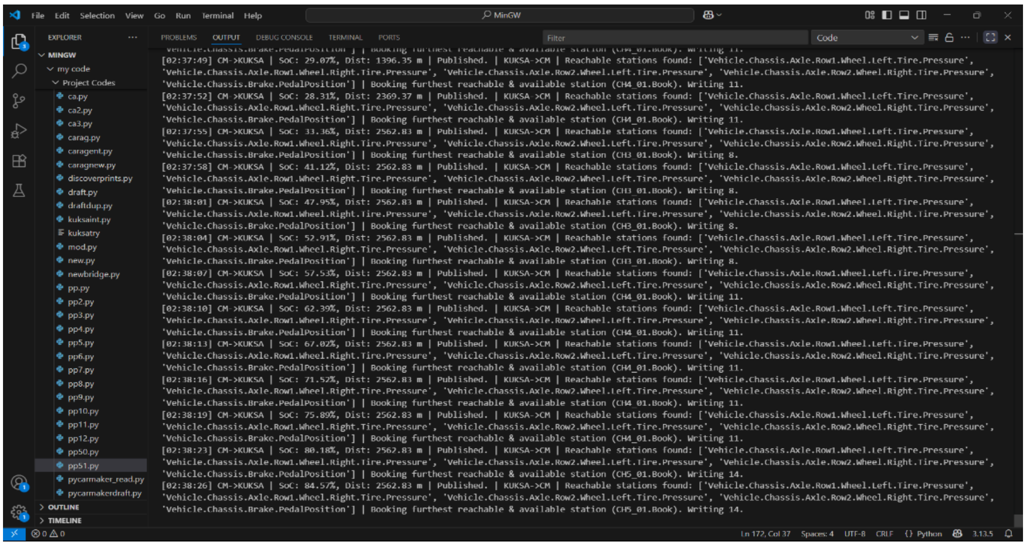


Fig 2. Car_Agent logs

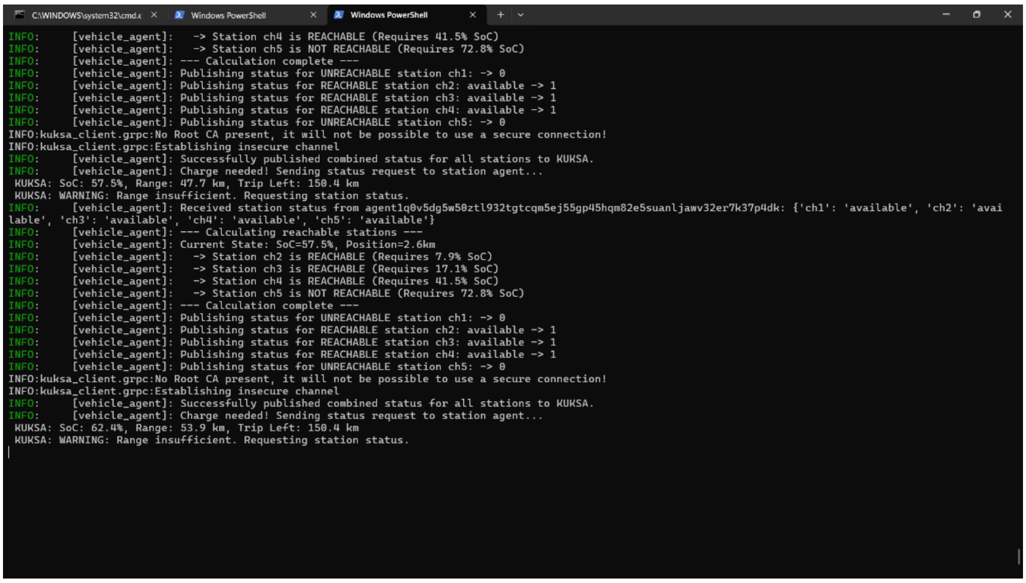
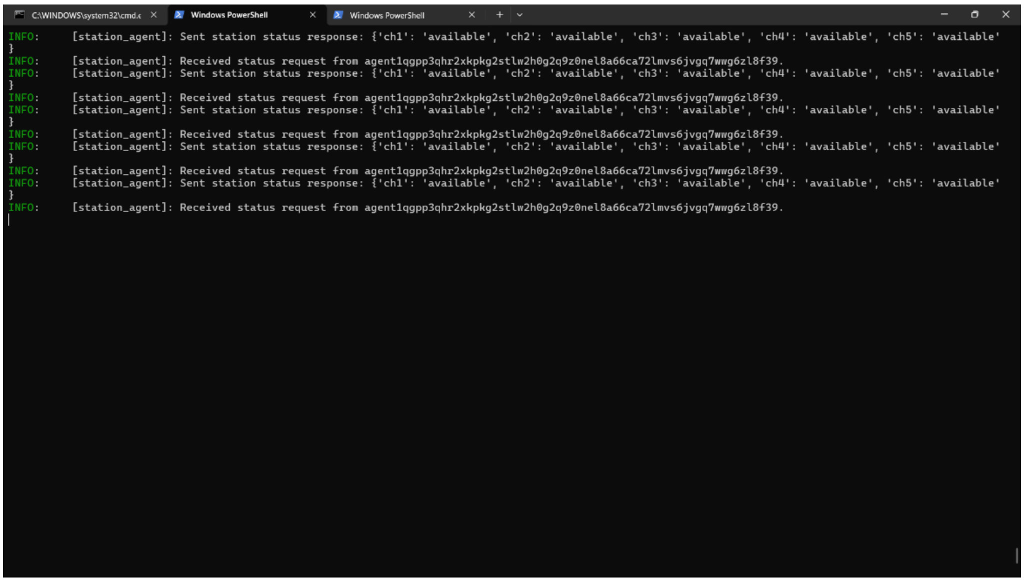


Fig 3. Station_agent Logs



Submitted on: 22/08/2025 **Submitted by:** Om Jokhir Athitya, Team Evolution